The Hong Kong Polytechnic University

Department of Electrical and Electronic Engineering

EIE4428 Multimedia Communications (2026/27)

Mini-Project: Remote Video Monitoring

Objectives

This is a system integration project, which is designed for students to learn fundamentals for data capture, compression, and transmission of multimedia data over networks.

Equipment

PCs/notebooks with camera.

Grouping

Each group should have two students.

Project Demonstration

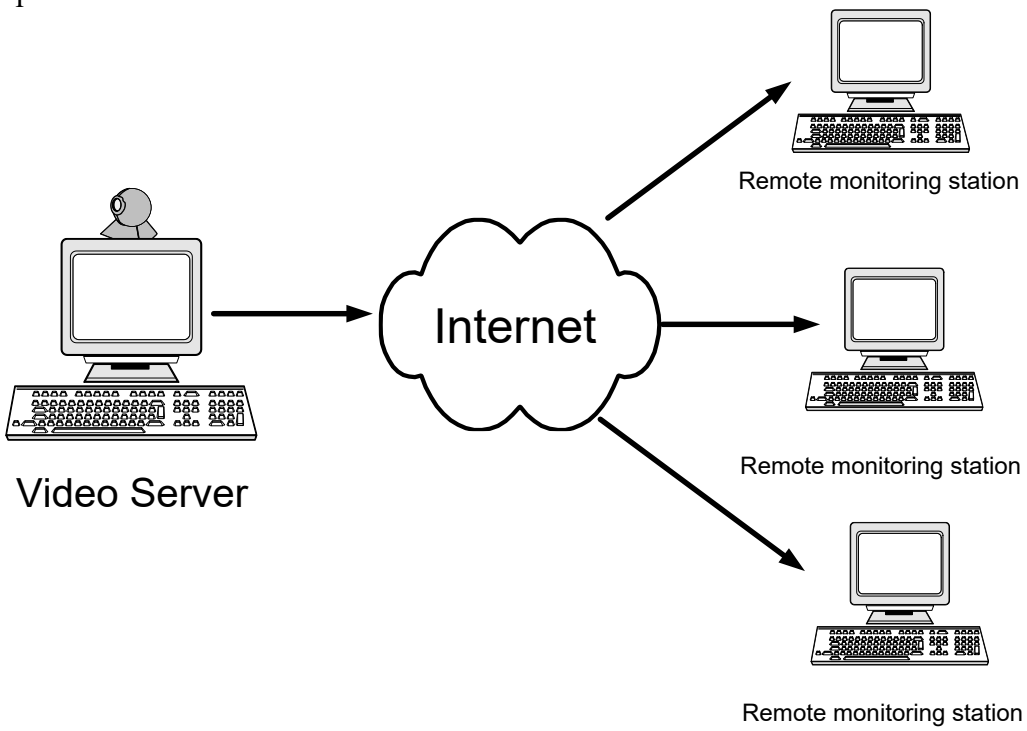
Week 13 (November 25 to November 28, 2026)

Report and Codes Submission Deadline

November 39, 2026 (Sunday) (One report per group)

Details of the Project

- Implement a remote video monitoring system, in which a user can monitor a place through a remote station (PC) using a standard Web browser like Chrome or Firefox or a standalone reception software.



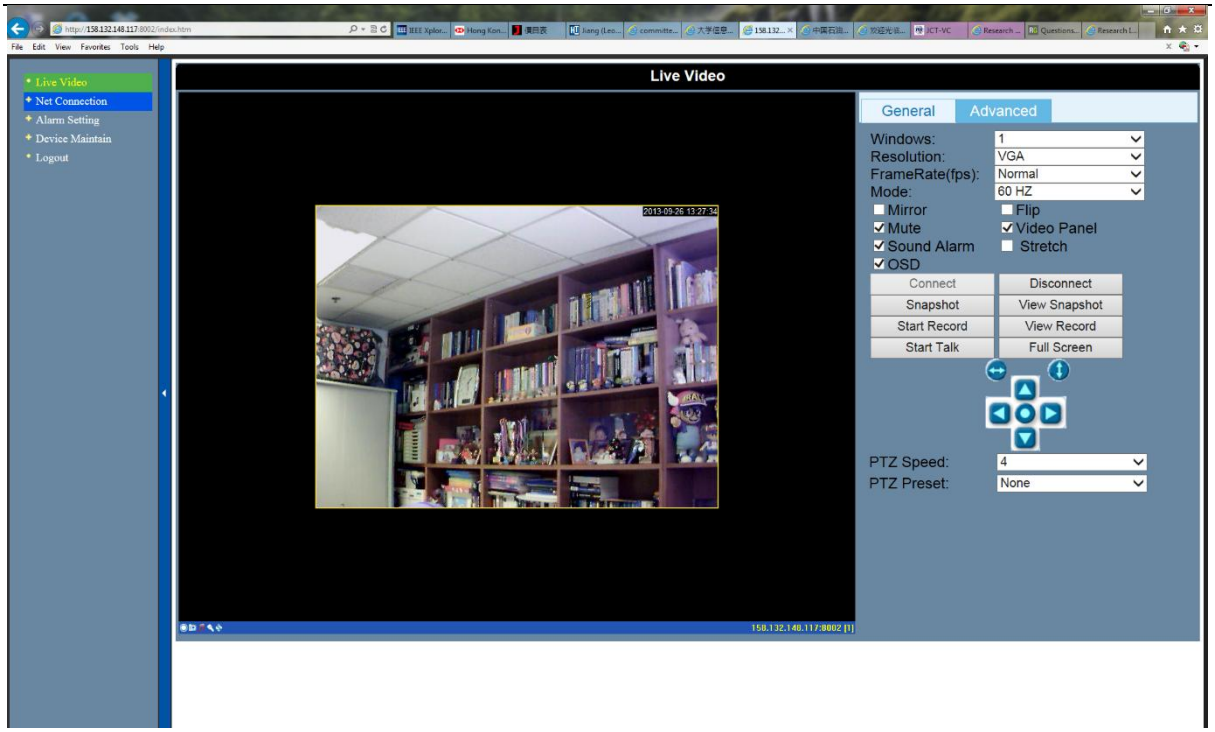
Remote video monitoring system

- The system will consist of a video server, an interface between video server and web, and a web page containing a link to the video server.

- The video server is used for video capture and compression. The video is first captured and each video frame is then compressed.

- At the remote station, users can use a standard web browser or a reception software installed at the client machine to access video content of the video server in real-time.

- A snap shot of a commercial remote video monitoring software at the client machine is shown below for your reference:



A snap shot of a commercial remote video monitoring software

- There are many approaches for implementations. One is to use a development tool like WebRTC. The other is to implement the system using your own codes.

- This system integration project can be completed using any available language and any development platform other than the tools mentioned below.

- Advanced features like timing information, motion detection (set alarm when a motion is detected in a static background), face detection, face recognition, video recording with timing information, unlimited concurrent access, audio signal, object tracking (tracking the position of the moving objects in the video), gesture recognition, as well as emotion recognition may be implemented.

Tools

- **WebRTC**

  WebRTC is a free, open project that provides browsers and mobile applications with Real-Time Communications (RTC) capabilities via simple APIs. More information can be found at:

  https://webrtc.org

  More information about the applications of WebRTC can be found in the following sites:

  https://web.dev/articles/webrtc-basics
  https://webrtc.github.io/samples/

- **Owncast**

  Owncast is a self-hosted live video and web chat server that allows you to stream your content to your audience without relying on third-party platforms. It has no forced ads, subscription fees, or viewer limits.

  https://owncast.online/

- **Open Streaming Platform**

  An open-source, self-hosted video streaming and sharing platform built on NGINX RTMP that handles both live broadcasts and pre-recorded video libraries.

  https://github.com/Open-Streaming-Platform/open-streaming-platform

- **Red5 Media Server**

  An open-source streaming server designed for developers who want to build custom live streaming and interactive video applications.

  https://www.red5.net/red5-media-server/

Assessment

- Each group should hand in a written report summarizing how you have implemented the system together with a zip file containing all source codes and executable files and upload via Canvas on or before **November 29, 2026 (Sunday).**

- Each group should demonstrate the system to the lecturer during the demonstration sessions, which will be held in **Week 13** (November 25 to November 28, 2026).

  The demonstration sessions will be scheduled around the week before the final submission.

- Marks will be allocated in consideration of the following aspects:

  - The functionality of the system;
  - The design of user interface;
  - The stability of the system;
  - The report of the project

  In general, more customizations can be made, more marks will be given.

- A satisfactory grade will be given to students who can achieve the following:
  - A workable and stable system that the user can view the video from a remote PC
  - A simple but user friendly user interface
  - An acceptable video quality
  - A report clearly showing the details of the implementation and the division of effort among group members.

- A better grade can be obtained if the students' work contains more advanced monitoring functions as mentioned before.